

Prajwal Devaraja

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Professional Summary

M.Sc. graduate in mechanical engineering with a strong foundation in technical software development and a proven ability to solve complex, interdisciplinary problems. Experienced in designing and implementing web APIs, IoT applications, and secure backend architectures. Proficient in Python, Django, Flask, and C#, with expertise in data visualisation, computational modelling, and relational database design. Demonstrated success in research and innovative projects, including digital twin technology and custom CA development. Adept at working under minimum supervision, with a focus on delivering innovative and user-centric solutions.

Skill

- **Programming:** Python, C#, C++, Html, Css, Javascript, Postgresql, SQL, Arduino, FreeRTOS (ESP32).
 - **Concepts:** Convex Hull, Mesh analysis, K-Means algorithm, r- tree, Object Oriented Prog. and Fourier Transform.
 - **Framework:** Django, Flask, Asp.net.
 - **Source Control:** Git Repository, familiar with AWS concepts.
 - **Visualisation:** Matplotlib, Plotly, Plotly.js.
 - **IoT & Security:** MQTT, Custom Certificate Authority, TLS Handshake.
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Work Experience

Freelancer

Xolcano Technologies Pvt Ltd

Chikkamagaluru, India

Jul 2024 to Present

Python, Django, Flutter, paho mqtt, postgresql

Designed and implemented a robust web API interface for IoT applications, ensuring scalability and modularity.

- Research and recommended an optimised tech stack in accordance to client requirements.
- Architected a secure and scalable backend with focus on separation of concerns and modular design.
- Implemented MQTTS (with TLS handshake) and integrated Django signals with a Flutter frontend for real-time data visualisation.
- Created a custom certificate authority to generate proprietary IoT device certificates, ensuring enhanced security and compliance.

Research Associate

University of Sheffield

Sheffield, UK

Jun 2024 to Jul 2024

Python, Flask, html, css, postgresql, docker, git, libraries: numpy, r-tree, plotly(frames, sliders)

Research and development of a live proximity and clearance tracking simulator for the digital twin.

- Efficient customer requirement analysis, planning a quick and effective project plan.
- A novel use of **r-tree** for tracking CAD data and faster calculation of clearance.
- Effectively utilised Plotly frames and sliders to develop an ergonomic visualisation for the simulator.

Research Associate

University of Sheffield

Sheffield, UK

Nov 2022 to Sep 2023

Python, Flask, html, css, plotly(frames, sliders), postgresql, git, libraries: numpy, pyMapdl, plotly(frames, sliders)

A continuation of Masters research project, the job role involved integration of **FEA in Digital twins**. Following a thorough literature review some of the key accomplishments were:

- Development of flask based web application with visualisation(**plotly**) and **interactive user interface**.
- Learning and utilisation of pyMAPDL(Ansys) for **digital simulation** and **gmsb** for mesh generation.
- Development of data reduction model of CAD, by utilising K-means algorithm and Fourier Transform.
- Tested the application against two real world examples - **3-storey structure and Aircraft**.
- Collaborated proactively and communicated ideas with the team and stakeholders.

Freelancer

Xolcano Technologies Pvt Ltd

Provided technical advisory for mechanical projects. Gained hands-on experience in circuit board design using EasyEDA, and successfully designed and implemented op amp amplifiers.

Chikkamagaluru, India

Jan 2020 to Jun 2021

Education and Qualifications

MSc Advanced Mechanical Engineering Merit(2:1), IMechE accredited

University of Sheffield

Sheffield, UK

Sep 2021 to Sep 2022

- **Key Modules:** **Advanced Dynamics**, Industrial Application of FEA, Exp. And valid computer models (Solids, Fluids), **Aviation Safety and Aero elasticity**, Design and Manufacture of Composite materials.
- **Mini Project:** "Assessment of implementing carbon fibre in public transport".
- **Research Project:** Developing an interactive CAD interface for a digital twin.

BE Mechanical Engineer - 1st class with Distinction

Adichunchanagiri Institute of Technology (VTU affiliated)

Chikkamagaluru, India

Aug 2012 to Aug 2016

- **Key Modules:** Mechanical Vibrations, Fluid Dynamics, FEA, Hydraulics and Pneumatics, Operations Research, CIM and Automated Lab, Kinematics, Heat transfer/Thermodynamics and Design of machines.
 - **Project:** "**Design and Fabrication of SAKAL Pot Maker**" – Machine to prepare seedling pots.
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Projects

Digital Twin Prototype(MSc Project)

Dec 2021 to Sep 2022

Developed dynamic **CAD interface** for a **digital twin** web application. The entire application was built to accept user fed design data in STL format and produce a **3D equivalent system**. The project development involved:

- Coordinating with stakeholders to communicate the ideas and identify required changes.
- Using rotation matrix to reorient user fed STL data and visualise it using **matplotlib**.
- Using **convex hull** theory effectively to identify contact between assembly parts.
- Staging of the assembly based on the contacts and generation of equivalent 3 DOF systems.

Design and development of Sakal Pot Maker(Self-Employed)

Jan 2017 to Aug 2020

Independently conceptualised, designed and analysed the idea as a solution to the local planter's labour problem in preparing nursery pots.

- Developed lightweight chassis, well designed for working stress- strain conditions and price.
 - Successful **innovation of a gear system** for potential **cost reduction**.
 - Coordinated with local startups for parts and electronic system development.
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Volunteer works

- o Took up a **teaching support role** at Uni of Sheffield. Assisted students in solving vibration problems.
 - o Assisted student in modelling and conducting **modal analysis of a bridge prototype** at University.
 - o Participated actively at the **AIUK** showcase event. Communicated the concept of digital twins to attendees with diverse expertise effectively conveying our project objectives.
 - o Led my undergraduate team final year project. Identified the best qualities of each individual utilising efficient project management producing excellent results.
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Interests

Sketching cars and concept car designs during my free time. Passionate about innovation and development process involved in new product development.

Reference

Reference available upon request